Solutions to Third Problem Sheet
on
Complex Analysis
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Topics: Infinite Series
Power Series
Sequences of Functions

"A mathematician, like a painter or a poet, is a maker of patterns. If his patterns are more permanent than theirs, it is because they are made with ideas."
-G.H. Hardy

Problem 1
Use mathematical induction to prove that

$$\sum_{k=0}^{n} z^k = \frac{1-z^{n+1}}{1-z} \quad \text{for all } z \in \mathbb{C} \setminus \{1\}.$$

Solution 1
It is clear that the formula holds for $n = 0$ so let us suppose that it holds for some $m \in \mathbb{N}$ so that

$$\sum_{k=0}^{m} z^k = \frac{1-z^{m+1}}{1-z}$$

whence

$$\sum_{k=0}^{m+1} z^k = \frac{1-z^{m+1}}{1-z} + z^{m+1} = \frac{1-z^{m+1}+z^{m+1}-z^{m+2}}{1-z} = \frac{1-z^{m+2}}{1-z}.$$



Problem 2
Prove Theorem 4.3 (Properties of Series).

Solution 2
i) Suppose that $\sum a_n = A$ and $\sum b_n = B$, then given $\varepsilon > 0$ choose $N \in \mathbb{N}$ to be such that

$$\left|\sum_{k=0}^{n} a_k - A\right| < \frac{\varepsilon}{2} \quad \text{and} \quad \left|\sum_{k=0}^{n} b_k - B\right| < \frac{\varepsilon}{2} \quad \text{for} \quad n \geq N,$$

this way, for $n \geq N$ one has

$$\left|\sum_{k=0}^{n}(a_k+b_k)-(A+B)\right| \leq \left|\sum_{k=0}^{n}a_k-A\right|+\left|\sum_{k=0}^{n}b_k-B\right|<\varepsilon.$$

ii) This part is obvious for $c=0$ so let us assume that $c\neq 0$. Now, given $\varepsilon>0$ choose N to be such that

$$\left|\sum_{k=0}^{n}a_k-A\right|<\frac{\varepsilon}{|c|} \quad \text{for} \quad n\geq N$$

so that

$$\left|\sum_{k=0}^{n}ca_k-cA\right|=|c|\left|\sum_{k=0}^{n}a_k-A\right|<|c|\frac{\varepsilon}{|c|}=\varepsilon.$$

iii) Suppose that $\sum_0 a_k$ converges to A, then given $\varepsilon>0$, there exists some $N\in\mathbb{N}$ for which

$$\left|\sum_{k=0}^{n}a_k-A\right|<\varepsilon \quad \text{for} \quad n\geq N$$

this way,

$$\left|\sum_{k=0}^{n}a_k-A\right|=\left|\sum_{k=0}^{m}a_k+\sum_{k=m}^{n}a_k-A\right|=\left|\sum_{k=m}^{n}a_k-\left(A-\sum_{k=0}^{m}a_k\right)\right|<\varepsilon$$

so that it is readily seen

$$\sum_{k=m}^{\infty}a_k=\sum_{k=0}^{\infty}a_k-\sum_{k=0}^{m-1}a_k$$

iv) This is a straightforward result of the identification $j=n-k$.

Problem 3

Show that if $X=(x_n)$ and $Y=(y_n)$ are real sequences and if for some $K\in\mathbb{N}$ we have

$$0\leq x_n\leq y_n \quad \text{for} \quad n\geq K$$

then:

i) the convergence of $\sum y_n$ implies the convergence of $\sum x_n$.

ii) the divergence of $\sum x_n$ implies the divergence of $\sum y_n$.

Solution 3

i) With the given assumption, suppose further that $\sum y_n$ converges, then by the Cauchy criterion, for each $\varepsilon>0$ there is $H_\varepsilon\in\mathbb{N}$ such that

$$|y_{n+1}+\cdots+y_m|=y_{n+1}+\cdots+y_m<\varepsilon \quad \text{for} \quad m>n\geq H_\varepsilon,$$

and thus for $m>\sup\{K,H_\varepsilon\}$ we have

$$0\leq x_{n+1}+\cdots+x_m\leq y_{n+1}+\cdots+y_m<\varepsilon$$

so the Cauchy criterion yields that $\sum x_n$ is also convergent, this proves i).

ii) Take the contrapositive of i).

Problem 4

If $\lim(x_n) = x > 0$, show that there exists a natural number K such that if $n \geq K$, then $\frac{1}{2}x < x_n < 2x$. Use this result to prove the **Limit comparison test**: If $X = (x_n)$ and $Y = (y_n)$ are strictly positive sequences and if

$$r := \lim\left(\frac{x_n}{y_n}\right)$$

exists in $\mathbb{R}$, then

i) if $r \neq 0$, then $\sum x_n$ is convergent if and only if $\sum y_n$ is convergent.

ii) If $r = 0$ and if $\sum y_n$ is convergent, then $\sum x_n$ is convergent.

Solution 4

Suppose that $\lim(x_n) = x > 0$, then given $\varepsilon = x/2$ there exists $K_\varepsilon \in \mathbb{N}$ such that $|x_n - x| < x/2$, or, equivalently, $x - x/2 < x_n < x + x/2$ and since $3x/2 < 2x$, the conclusion is now immediate. Next, suppose that r exists in $\mathbb{R}$, then for $r \neq 0$ we have that $r > 0$ since X, Y are strictly positive sequences and hence by the previous result we know that there is some K such that $r/2 < x_n/y_n < 2r$ for $n \geq K$, and the inequality is equivalent to

$$\frac{1}{2}ry_n < x_n < 2ry_n$$

from which one can conclude (by the comparison test) that $\sum x_n$ is convergent if and only if $\sum y_n$ is. For ii) we have that if $r = 0$ then for $\varepsilon = 1$ there is some K for which $0 < |x_n/y_n| < 1$, or, equivalently, $0 < x_n \leq y_n$ so that the comparison test yields the desired conclusion. ✡

Problem 5

Let $\sum_{n=0}^{\infty} a_n$ and $\sum_{n=0}^{\infty} b_n$ be two series in $\mathbb{C}$ converging to A and B respectively. The **Cauchy product** of these two series is defined as the series $\sum_{n=0}^{\infty} c_n$, where

$$c_n = \sum_{k=0}^{n} a_k b_{n-k}.$$

Show that if one of the series $\sum x_n$, $\sum y_n$ converges absolutely, then their Cauchy product converges to AB, this result is known as **Merten's theorem**. Further, consider the series

$$\sum_{n=0}^{\infty} \frac{(-1)^n}{\sqrt{n+1}},$$

which converges by the alternating series test (Leibniz criterion) but not absolutely. Show that the n-th term of its Cauchy product with itself is

$$c_n = \sum_{k=0}^{n} \frac{(-1)^n}{\sqrt{(k+1)(n-k+1)}}$$

and verify that $|c_n| \geq 1$ for all n, so that $c_n \not\to 0$. Conclude that the Cauchy product of two conditionally convergent series need not converge.

Solution 5

Let c_n be defined as before and thus without loss of generality, suppose that $\sum a_n$ is the absolutely convergent series. This way, we have the assumptions that $a_n \to A$, $b_n \to B$, $|a_n| \to M$, then

define

$$s_n^a = \sum_{j=0}^{n} a_j, \quad s_n^b = \sum_{j=0}^{n} b_j$$

so $s_n^a \to A$ and $s_n^b \to B$, and we can write $AB = (A + s_n^a - s_n^a)B = (A - s_n^a)B + s_n^a B$. On the other hand, we can also define

$$u_n = \sum_{j=0}^{n} c_j = c_0 + c_1 + \cdots + c_n = a_0b_0 + (a_0b_1 + a_1b_0) + \cdots + (a_0b_n + a_1b_{n-1} + \cdots + a_nb_0)$$

$$= a_0(b_0 + b_1 + \cdots + b_n) + a_1(b_0 + b_1 + \cdots + b_{n-1}) + \cdots + a_nb_0 = \sum_{j=0}^{n} a_j \left(\sum_{k=0}^{n-j} b_k \right) = \sum_{j=0}^{n} a_j s_{n-j}^b,$$

and this way,

$$|u_n - AB| = \left| \sum_{j=0}^{n} a_j s_{n-j}^b - (A - s_n^a)B - s_n^a B \right| = \left| \sum_{j=0}^{n} a_j s_{n-j}^b - (A - s_n^a)B - \sum_{j=0}^{n} a_j B \right|$$

$$= \left| \sum_{j=0}^{n} a_j (s_{n-j}^b - B) - (A - s_n^a)B \right| \leq \sum_{j=0}^{n} |a_j| \left| s_{n-j}^b - B \right| + |A - s_n^a||B| \qquad (*)$$

but given the hypotheses we can find $N_A, N_B \in \mathbb{N}$ such that

$$|A - s_n^a| < \frac{\varepsilon}{3(|B| + 1)} \quad \text{for} \quad n \geq N_A,$$

and

$$|B - s_n^b| < \frac{\varepsilon}{3(|A| + 1)} \quad \text{for} \quad n \geq N_B,$$

from which we get that the second term in $(*)$ is strictly less than $\varepsilon/3$. Moreover, if $n \geq N_B$, then we can split the first term into

$$\sum_{j=0}^{n-N_B} |a_j| \left| s_{n-j}^b - B \right| + \sum_{j=n-N_B+1}^{n} |a_j| \left| s_{n-j}^b - B \right|. \qquad (**)$$

For the first term in $(**)$ we have that $n - j \geq N_B$ so that

$$\sum_{j=0}^{n-N_B} |a_j| \left| s_{n-j}^b - B \right| < \sum_{j=0}^{n-N_B} |a_j| \frac{\varepsilon}{3(|A| + 1)} = \frac{\varepsilon}{3(|A| + 1)} \sum_{j=0}^{n-N_B} |a_j| < \frac{\varepsilon |A|}{3(|A| + 1)} < \frac{\varepsilon}{3},$$

whereas for the second term in $(**)$ we see that since $s_{n-j} \to B$ as $n \to \infty$, then the terms are bounded, i.e., there is some $K > 0$ for which $|s_m^b - B| \leq K$ for all $m \in \mathbb{N}$ and since the series $\sum |a_n|$ is convergent, then by the nth term test we have that $|a_n| < \varepsilon/3KN_B$ for $n \geq N_M$ thus if we also assume that $n \geq N_B + N_M$ we must have $j > N_M$ (because the first index j is $n - N_B + 1 \geq N_M + 1 > N_M$) so each $|a_j| < \varepsilon/3N_BK$, but there are exactly $n - (n - N_B + 1) + 1 = N_B$ terms in the sum and hence

$$\sum_{j=n-N_B+1}^{n} |a_j| \left| s_{n-j}^b - B \right| \leq K \sum_{j=n-N_B+1}^{n} |a_j| < KN_B \frac{\varepsilon}{3N_BK} = \frac{\varepsilon}{3}.$$

In order to satisfy the assumption made in the last step ($n \geq N_B + N_M$) and in the first step ($n \geq N_B$) it is enough to choose $n \geq N_B + N_M$, and in order to bound the second term in $(*)$ we need $n \geq N_A$ so by letting $N = \max(N_A, N_B + N_M)$ we have that if $n \geq N$, then

$$|u_n - AB| < \sum_{j=0}^{n-N_B} |a_j| \left| s_{n-j}^b - B \right| + \sum_{j=n-N_B+1}^{n} |a_j| \left| s_{n-j}^b - B \right| + |A - s_n^a||B| < \frac{\varepsilon}{3} + \frac{\varepsilon}{3} + \frac{\varepsilon}{3} = \varepsilon.$$

We therefore conclude that the Cauchy products converge to

$$\sum_{j=0}^{\infty} c_j = \sum_{j=0}^{\infty} a_j \sum_{j=0}^{\infty} b_j.$$

Now, let $a_n = b_n = (-1)^n/\sqrt{n+1}$ so that

$$c_n = \sum_{k=0}^{n} a_k b_{n-k} = \sum_{k=0}^{n} \frac{(-1)^k}{\sqrt{k+1}} \frac{(-1)^{n-k}}{\sqrt{n-k+1}} = \sum_{k=0}^{n} \frac{(-1)^n}{\sqrt{(k+1)(n-k+1)}},$$

this way,

$$|c_n| = \sum_{k=0}^{n} \frac{1}{\sqrt{(k+1)(n-k+1)}},$$

and by the AM-GM inequality we have

$$\sqrt{(k+1)(n-k+1)} \leq \frac{k+1+n-k+1}{2} = \frac{n+2}{2} \implies \frac{1}{\sqrt{(k+1)(n-k+1)}} \geq \frac{2}{n+2},$$

so that

$$|c_n| \geq \sum_{k=0}^{n} \frac{2}{n+2} = \frac{2(n+1)}{(n+2)},$$

and defining $\ell_n = 2(n+1)/(n+2)$ we see that (ℓ_n) is increasing since

$$\ell_{n+1} - \ell_n = \frac{2(n+2)}{(n+3)} - \frac{2(n+1)}{(n+2)} = \frac{2}{(n+2)(n+3)} \geq 0$$

and given that $\ell_0 = 1$, we conclude that $|c_n| \geq \ell_n \geq 1$ for all n. Given that $|c_n| \geq 1$ for all n, we conclude that $\lim(c_n) \neq 0$ so that (using the nth term test) the series $\sum c_n$ is divergent. ✡

Problem 6

Let $\sum a_n(z-z_0)^n$ be a power series with radius of convergence R given by Hadamard's formula. Then show that

$$R = \lim \left| \frac{a_n}{a_{n+1}} \right|$$

provided that this limit exists or equals ∞.

Solution 6

Suppose that $\lim |a_n/a_{n+1}|$ exists and equals L, then consider $w \equiv z - z_0 < r < L$. Since $\lim |a_n/a_{n+1}| = L$ then for $L - r/2 > 0$ there exists an integer N such that

$$\left| \left| \frac{a_n}{a_{n+1}} \right| - L \right| < L - \frac{r}{2} \iff \frac{r}{2} - L < \left| \frac{a_n}{a_{n+1}} \right| - L < L - \frac{r}{2} \implies \left| \frac{a_n}{a_{n+1}} \right| > \frac{r}{2} > r \quad \text{for} \quad n \geq N.$$

Letting $A = |a_N| r^N$ and noticing that

$$|a_{N+k}| r^{N+k} = |a_{N+k}| r^{N+k} \frac{|a_{N+k-1}| r}{|a_{N+k-1}| r} = |a_{N+k-1}| r^{N+k-1} \frac{|a_{N+k}|}{|a_{N+k-1}|} r < |a_{N+k-1}| r^{N+k-1}$$

for $k \geq 1$, we get $|a_n| r^n \leq A$ for all $n \geq N$. Then $|a_n w^n| = |a_n| r^n (|w|/r)^n \leq A(|w|/r)^n$ and since $|w| < r$, the geometric series $\sum (|w|/r)^n$ is convergent so that by the comparison test we get that $\sum |a_n w^n|$ is convergent too. Since $r < L$ was arbitrary and $\sum a_n w^n$ converges absolutely for

$|w| < r$, if we suppose $R < L$ with R the radius of convergence, then take $r, |w|$ to be such that $R < |w| < r < L$, then $\sum |a_n w^n|$ converges, but this is a contradiction since $|w| > R$ implies that the series $\sum |a_n w^n|$ diverges. We therefore conclude that $R \geq L$. If $L = \infty$ this automatically yields $R = \infty = L$.

If L is finite, we proceed similarly, consider $|w| > r > L$, using the same reasoning as before (now take $r/2 - L$), we have that there exists an integer M such that $|a_n/a_{n+1}| < r$ for $n \geq M$. Setting $B = |a_M| r^M$ we have that

$$|a_{M+k}| r^{M+k} > |a_{M+k-1}| r^{M+k-1}$$

so that $|a_n r^n| \geq B$ for all $n \geq M$. Then $|a_n w^n| = |a_n| r^n (|w|/r)^n \geq B(|w|/r)^n$ for $n \geq M$ and since $|w| > r$, we get $\lim |a_n w^n| = \infty$. Since $r > L$ was arbitrary and $\sum a_n w^n$ diverges for $|w| > r$, suppose $R > L$, then we may take $L < r < |w| < R$, whence $|w| > r$ and thus the series $\sum a_n w^n$ diverges but the definition of R implies that (since $|w| < R$) $\sum a_n w^n$ converges absolutely, a contradiction. We therefore conclude that we must have $R \leq L$. Using the first part we have

$$L \leq R \leq L,$$

from which the conclusion follows readily.

Problem 7

Use Merten's theorem to show that

$$\sum_{k=0}^{\infty} \frac{z^k}{k!} \sum_{k=0}^{\infty} \frac{w^k}{k!} = \sum_{k=0}^{\infty} \frac{(z+w)^k}{k!}.$$

Solution 7

Using the previous exercise we have that the radius of convergence of each series is given by

$$R = \lim \left| \frac{a_n}{a_{n+1}} \right| = \lim \frac{(k+1)!}{k!} = \lim(k+1) = \infty.$$

Since both the series converge absolutely for all $z, w \in \mathbb{C}$ we can apply Merten's theorem to get that the Cauchy product

$$\sum_{n=0}^{\infty} c_n = \sum_{n=0}^{\infty} \sum_{k=0}^{n} \frac{z^k}{k!} \frac{w^{n-k}}{(n-k)!},$$

but notice how

$$\frac{(z+w)^n}{n!} = \frac{1}{n!} \sum_{k=0}^{n} \binom{n}{k} z^n w^{n-k} = \frac{1}{n!} \sum_{k=0}^{n} n! z^n w^{n-k} \frac{1}{k!(n-k)!} = \sum_{k=0}^{n} \frac{z^n w^{n-k}}{k!(n-k)!},$$

so that

$$\sum_{n=0}^{\infty} c_n = \sum_{n=0}^{\infty} \frac{(z+w)^n}{n!}.$$

Problem 8

Let (f_n) be a sequence of complex functions on $\Omega \subseteq \mathbb{C}$. Suppose that for every non-negative integer n, there exists $M_n > 0$ such that $|f_n| \leq M_n$ for all $z \in \Omega$ and such that $\sum M_n$ is convergent. Hence show that the series $\sum f_n$ is uniformly convergent throughout. This result is called the **Weierstrass criterion** for complex series of functions.

Solution 8

Let $f_n : \Omega \to \mathbb{C}$. Suppose that for each $n \in \mathbb{N}$ there exist a positive number M_n such that $|f_n(z)| \leq M_n$ for $z \in \Omega$, then for $m > n$ we have

$$|f_{n+1}(z) + \cdots + f_m(z)| \leq M_{n+1} + \cdots + M_m \quad \text{for} \quad z \in \Omega,$$

so that the Cauchy criterion for series and the convergence of $\sum M_n$ yields the desired result. ✡